Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Virology / Viral Safety	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ANOVA, analysis of variance · CCD, central-composite design · CPP, critical process parameter · CQA, critical quality attribute · DoE, design of experiments · DS, drug substance · GPP, general process parameter · HMW, high-molecular-weight (aggregate) · ICH, International Council for Harmonisation · icIEF, imaged capillary isoelectric focusing · LRF, log-reduction factor · MSAT, Manufacturing Science and Technology · MVM, minute virus of mice · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · TCID50, 50% tissue-culture infectious dose · WC-CPP, well-controlled CPP · XMuLV, xenotropic murine leukemia virus.

1 Purpose and scope

This Process Characterization Plan (**PCP-006**) defines the Stage 1 (Process Design) characterization of the A-Mab low-pH viral inactivation step (Step 6), the dedicated viral-clearance operation in which the Protein A eluate is held at low pH to inactivate enveloped viruses. The plan specifies the objectives, the attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and a design space established.

Scope. The commercial-scale low-pH inactivation step, characterized on a qualified small-scale inactivation model with XMuLV spiking studies. The upstream Protein A step and the downstream polishing and virus-filtration steps are addressed only where they define the feed to, or complement the clearance of, this step; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-006**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012); viral clearance is addressed under ICH Q5A (International Council for Harmonisation 2023a); and the lifecycle approach follows the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. establish the quantitative relationships between the inactivation parameters and the XMuLV log-reduction, the aggregate and the acidic charge variants;
2. identify the significant main effects, two-factor interactions and curvature;
3. define the multivariate design space over which the enveloped-virus inactivation is assured while aggregate remains within its limit;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the viral-safety control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Low-pH viral inactivation is the dedicated inactivation step of the A-Mab purification train: the Protein A eluate, already at low pH from the acidic elution, is titrated to the target inactivation pH, held for a defined time at controlled temperature, and then neutralized before the next step (CMC Biotech Working Group 2009; International Council for Harmonisation 2023a). Low pH disrupts the lipid envelope of enveloped viruses and is a robust clearance mechanism for the endogenous retrovirus-like particles modelled by XMuLV; it is ineffective against the non-enveloped parvovirus model MVM, whose clearance is provided orthogonally by anion exchange and small-virus filtration. Platform knowledge established that inactivation increases as pH is lowered and as hold time and temperature increase; that the acceptable pH window is narrow and high-consequence (above ~4.0 the inactivation is incomplete; below ~3.2 the product is at risk of acid-induced aggregation); and that aggregate can form during a prolonged or warm hold. This knowledge frames the attributes in scope and the prioritization of parameters.

4.2 Quality attributes in scope

The step sets the (cumulative) enveloped-virus clearance CQA (Table 4.1) and carries a product-quality risk of aggregate formation during the low-pH hold; acidic charge variants are followed as they can shift slightly during the hold. Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact × Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

Table 4.1: Viral-clearance CQA set by the low-pH inactivation step, with acceptance criterion and criticality (Tool #1 = Impact × Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

The XMuLV clearance CQA is expressed as the cumulative log-reduction across the process (16.7 log), to which this step is the single largest contributor; the aggregate is a High-criticality size-variant CQA (limit 5% HMW) that this step can increase and that cation exchange subsequently polishes.

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments and the risk flow of PDA TR 60, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential impact and assigns the study type accordingly: inactivation pH, hold time and temperature (credible impact to the enveloped-virus inactivation and to aggregate) are studied in the multivariate DoE; A-Mab concentration, expected to affect neither over a wide range, is studied univariately (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a small-scale inactivation model operated at the commercial-process equivalents for acidification, mixing, hold vessel geometry and temperature control (**SOP-2009**). Viral log-reduction will be determined in spiking studies in which a characterized XMuLV stock is added to the starting material and the infectious titre measured before and after the hold. Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and the output attributes — XMuLV log-reduction, aggregate and charge variants — to confirm the model is representative and suitable to support design-space and PAR claims (Parenteral Drug Association 2013; International Council for Harmonisation 2023a). The centre-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Operation

The Protein A eluate is titrated with acid to the target inactivation pH, mixed, and held for the defined hold time at controlled temperature; the solution is then neutralized to the load condition of the next step. Inactivation pH and temperature are controlled to set-point and the hold time is timed from attainment of the target pH. Buffers and the A-Mab concentration of the incoming pool are controlled to platform specifications and are treated as quality-controlled inputs, except for the concentration range assessed univariately (CMC Biotech Working Group 2009).

5.3 Analytical methods

The step responses will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

5.4 Sampling plan

Each DoE run will be sampled before and after the hold for the XMuLV infectious titre and at the end of the hold for aggregate and acidic charge variants; in-process pH, temperature and hold-time records will confirm the run conditions. Centre-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model. Significance will be assessed at $= 0.05$. Model adequacy will be judged by R^2, adjusted R^2, predicted R^2, the overall model F-test, and an ANOVA lack-of-fit test against the centre-point pure error. Consistent with the conservative treatment of viral clearance (International Council for Harmonisation 2023a), the enveloped-virus clearance claim will be taken from the demonstrated log-reduction rather than from an extrapolated model. The design space will be defined as the region of inactivation pH and hold time over which the XMuLV log-reduction is assured and aggregate remains within its limit, and commercial-scale capability of the cumulative XMuLV clearance will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Four parameters are in scope (Table 6.1). The characterization range of each factor ("Range studied") defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-006. The DoE factor coding for the three multivariate factors is given in Table 6.2.

Table 6.1: Low-pH inactivation parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Inactivation pH	pH	3.5	3.2–4	3.4–3.6	multivariate
Hold time	min	90	60–180	60–120	multivariate
Temperature	°C	21	15–25	19–23	multivariate
A-Mab concentration	g/L	20	5–35	5–35	univariate

Table 6.2: Design-of-experiments factor legend (coded A–C).

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

6.2 Screening design

Factor screening will use a two-level full factorial in the three multivariate factors (A–C) with three centre-point replicates, 11 runs in total. The full factorial estimates all three main effects and all three two-factor interactions without confounding. Three responses will be recorded for each run (XMuLV log-reduction, aggregate and acidic charge variants). The planned coded design is given in Appendix A.

6.3 Response-surface design

The three multivariate factors will be carried into a face-centred central-composite design with four centre-point replicates, 18 runs in total, enabling estimation of curvature in addition to main effects and interactions. The planned coded design is given in Appendix B.

6.4 Univariate assessment

A-Mab concentration will be evaluated one factor at a time across its characterization range for its effect on the XMuLV log-reduction and aggregate.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**);
2. the fitted response-surface model for the aggregate is adequate — target R^2 0.90, no significant lack of fit ($p > 0.05$ against the centre-point pure error), and residual diagnostics without systematic structure — and the XMuLV log-reduction model is adequate (no significant lack of fit), with the clearance claim taken conservatively from the demonstrated log-reduction rather than the model (International Council for Harmonisation 2023a); and
3. a design space exists over which the XMuLV log-reduction is assured and aggregate remains within its limit at the set-point and across each parameter's NOR.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts a CQA is a **CPP**; where the impact is reliably controlled within the design space (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; otherwise it is a **GPP**;
- consistent with the post-characterization FMEA convention, a parameter that affects a CQA with Severity 8, or that carries an initial (pre-mitigation) risk priority number > 72, is designated a CPP before mitigation by the control strategy.

The design space, classifications and PARs established here will justify the Stage 2 operating ranges and feed the viral-safety control strategy.

8 Data management and integrity

All raw data, DoE designs, analytical and infectivity-assay results and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and the derived design space will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-006.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-006
Analytical Development	Perform aggregate and charge-variant testing under the referenced AMV methods
Virology / Viral Safety	Execute the XMuLV spiking studies and infectivity assays; review clearance claims
Statistics	Review the design, the statistical analysis and the design-space definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-006**), reporting the executed designs, the significant effects, the design space, the parameter classification and the viral-clearance and aggregate outcomes. PCR-006 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ and viral-clearance validation campaigns.

11 Risks and assumptions

The plan assumes the SDM is representative of commercial scale (confirmed by qualification) and that the XMuLV stock and infectivity assay are qualified and in control. The principal risk is that the infectivity assay's inherent variability limits the precision of the modelled log-reduction; this is mitigated by taking the clearance claim conservatively from the demonstrated log-reduction, by controlling inactivation pH to a tight NOR, and by confirming the clearance in the dedicated viral-clearance validation (International Council for Harmonisation 2023a).

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table 1.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

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International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.

Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded two-level full-factorial design (factors A–C per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-006.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A–C; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0